MCQ solutions

1. **B.** The vector multiplying t is a direction vector; the first triple denotes a point on the line.
2. **D.** Take $t = 1$ satisfies all three coordinates simultaneously. A single correct coordinate does not is not enough.
3. **A.** In an orthonormal coordinate system, the coefficients of x, y, z give a normal vector. The term constant is not its third coordinate.
4. **C.** These two conditions characterise the projection. If M is in the plane, the zero vector works and $H = M$.
5. **B.** The new point is on the line, and 3 u is a non-zero direction vector collinear with u. The change $t = 2 + 3s$ runs through all $\mathbb{R}$.
6. **D.** No value of the parameter works: the line is strictly parallel to the plane.
7. **A.** Every point of the line satisfies the equation of the plane. The intersection is the whole line.
8. **C.** The same point can be reached for two different values of the parameters. Three coordinates must be checked.
9. **B.** The two constraints leave one free parameter; take $x = t$ imposes $y = 1 - t$ and $z = 2$.
10. **D.** The coefficient of y remains non-zero. An equation can still define a plane when a coefficient is zero.

True/false. False: they may be non-coplanar. False: they have collinear normal vectors but incompatible constants; the planes are strictly parallel. True: the normal line through M intersects the plane at a single point.

Solutions to the applications and guidance

Application 1. Let t=2+3s; substitution gives the three coordinates of the second parametrisation and runs through $\mathbb{R}$. For (1;3;2), t=1 satisfies the three coordinates.

Application 2. Equation $x+2y-z=1$. The proposed point gives $1+2-2=1$, so it belongs to the plane. Possible directions $(2;-1;0)$ and $(1;0;1)$, orthogonal to the normal and not collinear.

Application 3. d_1 intersects at (1;0;1); d_2 is strictly parallel; d_3 is included. Substitution gives t=1, 0=1 and 1=1, respectively.

Application 4. The equations are t=1+s, t=1-s and 0=s. So s=0, t=1, point (1;1;0). Solving two equations can produce candidates; only checking the third one is conclusive.

Application 5. $d_m : (t; mt; m+t)$. The equation becomes $2mt-m=1$. If m=0, it is impossible: the line strict parallel. Otherwise $t=(m+1)/(2m)$ provides a single point. To be perpendicular, $(1;m;1)$ would have to be collinear with (1;2;-1); the first coordinate forces a ratio of 1, incompatible with the third. No m.

Application 6. x=y=t and z=3-2t, so line $(t;t;3-2t)$, point (0;0;3), direction (1;1;-2). The point satisfies both equations; the direction vector is orthogonal to both normals (1;1;1) and (1;-1;0).

From the system to geometry

Have students predict the three cases from the figure, then read the equations 0=1 and 0=0 as useful results. To differentiate, provide the substitution to a student who has grasped the idea but gets lost in the calculation.

Teaching notes and solutions

Guidance for the teacher

Use two different parameters to compare two lines. Check the three coordinates, even though two are enough to compute candidates. A Cartesian equation alone generally describes a plane in space; it does not describe a line. Reuse the projection from chapter 11 for give meaning to solving systems.

Solutions to the activity

1. A point of d_1 has coordinates (t,t,t). It lies in the plane if and only if 3 $t=3$, i.e. $t=1$. The intersection is therefore the unique point $(1,1,1)$.

2. For a point of d_2, $x+y+z=3+t-t+0=3$ whatever t. All points of the line are in the plane: $d_2 \subset \mathcal{P}$.

3. For a point of d_3, $x+y+z=t-t+0=0$. The equation 0 $=3$ has no solution: the line is disjoint from the plane. Its direction vector $(1,-1,0)$ is orthogonal to the normal vector $(1,1,1)$; it is parallel to plane.

4. After substitution, we obtain respectively an equation with a single solution, an identity true for every value of the parameter, and a contradiction. These three cases correspond to an intersection at a point, inclusion and no intersection.

E1. $\overrightarrow{AB}=(2,1,-1)\neq\vec{0}$. The line therefore has the representation $(x,y,z)=(1,-1,2)+t(2,1,-1)$, $t\in\mathbb{R}$. The values $t=0$ and $t=1$ give back A and B.

E2. For $A(1,2,3)$, the condition $\overrightarrow{AM}\cdot(1,0,-2)=0$ is $(x-1)-2(z-3)=0$. After expanding, the plane has equation $x-2z+5=0$. The point A satisfies it.

E3. After substitution, $t+2(1+t)+(2-t)=8$, i.e. 2 $t+4=8$, so $t=2$. The unique intersection is $(2,3,0)$; check 2 $+2\times3+0=8$.

E4. The second equation gives $x=y$. Set $x=y=t$; the first becomes 2 $t+z=1$. The set of solutions is $(x,y,z)=(t,t,1-2t)=(0,0,1)+t(1,1,-2)$, a line.

E5. The normal vectors are collinear, the second being twice the first. But doubling 2 $x-y+z=3$ gives 4 $x-2y+2z=6$, incompatible with the right-hand side 5. The planes are therefore strictly parallel, and not intersecting.

Solutions to homework DM12

1. The coefficients x, y, z give the non-zero normal vector $\vec{n} = (1, 2, 2)$ in the orthonormal coordinate system.

2. Let us look for H on the normal through M : $H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We check $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. Hence H is indeed the projection, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.

3. The normal vectors $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the planes intersect in a line.
Since $x = y$, let $x = y = s$. The first equation gives 3 $s + 2z = 9$, i.e. $z = \frac{9-3s}{2}$.
The intersection is the line $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.

4. The point H a $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it does not belong to $\mathcal{Q}$ nor to the line of intersection.

5. The z-coordinate is the coordinate z. Setting it to zero requires $9 - 3s = 0$, therefore $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result agrees with the property of the projection: among all points of the plane, H is closest to M.

Solutions to the textbook or past-paper work

1. True: the line passes through B and its direction vector $(-2; -1; 3)$ is $-\frac{1}{2}\overrightarrow{AB}$.

2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.

3. False: for $k = -4$ and $s = \frac{5}{2}$, both representations give $(11; 12; -14)$. The lines intersect, so they are coplanar.

4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.